

AB-100 — Microsoft Agentic AI Business Solutions Architect:

Practice Questions & Answers with Explanations

DOMAIN 1 — Plan AI-Powered Business Solutions

Q1. A company wants to implement an AI-powered HR assistant that answers employee questions using internal HR policies stored in SharePoint. Which architecture is BEST suited?

- A. Traditional chatbot with static responses
- B. RAG architecture using Azure AI Search
- C. Hardcoded REST APIs only
- D. SQL stored procedures

Correct Answer:

- B. RAG architecture using Azure AI Search

Explanation:

RAG (Retrieval-Augmented Generation) allows the AI system to retrieve relevant HR documents from SharePoint using Azure AI Search and generate contextual responses dynamically.

Q2. Which Microsoft service is primarily used to build enterprise conversational AI agents with low-code capabilities?

- A. Azure Kubernetes Service
- B. Copilot Studio
- C. Azure Virtual Desktop
- D. Microsoft Intune

Correct Answer:

- B. Copilot Studio

Explanation:

Copilot Studio enables low-code/no-code enterprise AI agent creation with integrations into Power Platform and Microsoft services.

Q3. A company wants AI agents to collaborate together for procurement approvals, budget validation, and vendor verification. What architecture pattern should be used?

- A. Single monolithic chatbot
- B. Static workflow engine
- C. Multi-agent architecture
- D. File-based processing

Correct Answer:

- C. Multi-agent architecture

Explanation:

Multi-agent systems allow specialized agents to collaborate and delegate tasks.

Q4. Which Microsoft framework helps enterprises adopt AI solutions systematically?

- A. Azure Networking Framework
- B. Cloud Adoption Framework for Azure
- C. Azure Backup Framework
- D. Windows Deployment Framework

Correct Answer:

- B. Cloud Adoption Framework for Azure
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Q5. Which factor is MOST important during AI business ROI analysis?

- A. Monitor size
- B. GPU fan speed
- C. Business value and operational efficiency
- D. Laptop RAM

Correct Answer:

- C. Business value and operational efficiency
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Q6. Which scenario BEST fits Agentic AI?

- A. Static FAQ website
- B. Autonomous ticket routing and escalation
- C. PDF viewer
- D. Simple HTML form

Correct Answer:

B. Autonomous ticket routing and escalation

Q7. A company wants to reduce repetitive IT support tickets. Which AI solution is BEST?

- A. Manual email routing
- B. AI-powered IT helpdesk agent
- C. Spreadsheet tracking
- D. Offline documentation

Correct Answer:

B. AI-powered IT helpdesk agent

Q8. Which component stores enterprise vector embeddings for semantic search?

- A. Azure AI Search
- B. Paint
- C. Azure DNS
- D. Windows Defender

Correct Answer:

A. Azure AI Search

Q9. Which AI architecture pattern allows AI responses to be grounded on enterprise documents?

- A. ETL
- B. RAG
- C. VPN
- D. FTP

Correct Answer:

B. RAG

Q10. What is the PRIMARY purpose of prompt engineering?

- A. Reduce monitor brightness
- B. Optimize AI behavior and responses
- C. Format hard disks
- D. Restart Azure

Correct Answer:

B. Optimize AI behavior and responses

DOMAIN 2 — Design AI-Powered Business Solutions

Q11. Which Azure service provides enterprise GPT model hosting?

- A. Azure Backup
- B. Azure OpenAI Service
- C. Azure Monitor
- D. Azure Files

Correct Answer:

B. Azure OpenAI Service

Q12. Which feature allows AI systems to call external APIs dynamically?

- A. Blob Storage
- B. Function Calling
- C. Azure Firewall
- D. SMB

Correct Answer:

B. Function Calling

Q13. What is the purpose of embeddings in AI systems?

- A. Encrypt data
- B. Convert text into vector representations
- C. Compress videos
- D. Create virtual machines

Correct Answer:

B. Convert text into vector representations

Q14. Which Microsoft API is used for Outlook, Teams, and Calendar integrations?

- A. Microsoft Graph API
- B. JDBC API
- C. Kubernetes API
- D. Linux API

Correct Answer:

A. Microsoft Graph API

Q15. Which prompt type defines the overall AI behavior?

- A. User prompt
- B. System prompt
- C. SQL prompt
- D. Browser prompt

Correct Answer:

B. System prompt

Q16. Which architecture component is responsible for routing tasks between agents?

- A. GPU
- B. Orchestrator agent
- C. Power cable
- D. VPN

Correct Answer:

B. Orchestrator agent

Q17. What is the advantage of multi-agent AI systems?

- A. Reduced collaboration
- B. Specialized task handling
- C. No scalability
- D. No automation

Correct Answer:

B. Specialized task handling

Q18. Which technology enables event-driven enterprise workflows?

- A. Paint
- B. Power Automate
- C. BIOS
- D. CMD

Correct Answer:

B. Power Automate

Q19. Which AI technique improves enterprise document retrieval quality?

- A. Semantic search
- B. BIOS update
- C. Disk cleanup
- D. DHCP

Correct Answer:

A. Semantic search

Q20. Which design principle is MOST important for enterprise AI security?

- A. Open public access
- B. Least privilege access
- C. Shared admin passwords
- D. No monitoring

Correct Answer:

B. Least privilege access

DOMAIN 3 — Deploy AI-Powered Business Solutions

Q21. Which Azure identity service secures enterprise AI systems?

- A. Azure Entra ID
- B. Paint
- C. Notepad
- D. BIOS

Correct Answer:

A. Azure Entra ID

Q22. What is RBAC used for?

- A. Video editing
- B. Access control
- C. Audio streaming
- D. DNS management

Correct Answer:

B. Access control

Q23. Which feature helps prevent secret exposure in Azure services?

- A. Managed identities
- B. Task Manager
- C. Recycle Bin
- D. DHCP

Correct Answer:

A. Managed identities

Q24. Which AI security risk involves manipulating prompts maliciously?

- A. VPN tunneling
- B. Prompt injection
- C. DHCP failure
- D. RAM corruption

Correct Answer:

B. Prompt injection

Q25. Which Azure service helps monitor token usage and AI performance?

- A. Azure Monitor
- B. Paint
- C. WordPad
- D. Hyper-V

Correct Answer:

A. Azure Monitor

Q26. What does high availability ensure?

- A. Slower performance
- B. Minimal downtime
- C. Increased outages
- D. Reduced scalability

Correct Answer:

B. Minimal downtime

Q27. Which strategy improves AI cost optimization?

- A. Increase token usage unnecessarily
- B. Use smaller optimized models
- C. Disable monitoring
- D. Duplicate requests

Correct Answer:

B. Use smaller optimized models

Q28. Which deployment strategy reduces production deployment risk?

- A. Blue-green deployment
- B. Manual copy-paste
- C. Disable backups
- D. Shared passwords

Correct Answer:

A. Blue-green deployment

Q29. Which compliance regulation focuses on data privacy?

- A. GDPR
- B. HDMI
- C. DHCP
- D. FAT32

Correct Answer:

A. GDPR

Q30. Which architecture pattern improves disaster recovery?

- A. Single-region deployment
- B. Multi-region deployment
- C. Local-only storage
- D. Manual backups only

Correct Answer:

B. Multi-region deployment

Scenario-Based Questions

Q31. A company wants AI-generated responses grounded on enterprise documents while preventing hallucinations. Which solution should be implemented?

- A. Static HTML
- B. RAG architecture with Azure AI Search
- C. VPN gateway
- D. DNS failover

Correct Answer:

- B. RAG architecture with Azure AI Search
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Q32. An enterprise wants AI approval workflows integrated into Teams and Outlook. Which Microsoft service is BEST?

- A. Power Automate
- B. Azure DNS
- C. Hyper-V
- D. Windows Media Player

Correct Answer:

- A. Power Automate
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Q33. Which AI governance principle ensures humans remain accountable for AI decisions?

- A. Transparency
- B. Accountability
- C. Scalability
- D. Encryption

Correct Answer:

- B. Accountability
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Q34. Which architecture BEST supports long-running enterprise workflows?

- A. Stateless chatbot
- B. Stateful workflow system

- C. Offline spreadsheet
- D. FTP server

Correct Answer:

- B. Stateful workflow system
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Q35. Which Microsoft tool is MOST suitable for low-code AI workflow automation?

- A. Power Automate
- B. VMware
- C. BIOS
- D. IIS only

Correct Answer:

- A. Power Automate
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Advanced Questions

Q36. Which component stores conversational memory in enterprise AI systems?

- A. Stateful workflow layer
- B. Printer driver
- C. Mouse driver
- D. Firewall cable

Correct Answer:

- A. Stateful workflow layer
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Q37. Which Azure feature protects AI APIs using centralized security policies?

- A. Azure API Management
- B. MS Paint
- C. Windows Update
- D. SMB

Correct Answer:

- A. Azure API Management
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Q38. Which monitoring metric directly impacts Azure OpenAI cost?

- A. Token usage
- B. Screen resolution
- C. Keyboard layout
- D. USB speed

Correct Answer:

- A. Token usage
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Q39. Which AI pattern allows agents to collaborate hierarchically?

- A. Parent-child agent architecture
- B. DHCP architecture
- C. BIOS architecture
- D. FTP architecture

Correct Answer:

- A. Parent-child agent architecture
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Q40. Which Microsoft technology enables adaptive cards in Teams integrations?

- A. Copilot Studio
- B. Windows Media Center
- C. FTP Client
- D. Registry Editor

Correct Answer:

- A. Copilot Studio
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HIGH-DIFFICULTY AB-100 QUESTIONS

Q41. A multinational company requires AI systems to comply with regional data residency requirements. Which architecture consideration is MOST important?

- A. GPU size
- B. Regional AI deployments
- C. Local printer drivers
- D. Browser extensions

Correct Answer:

- B. Regional AI deployments

Q42. Which AI deployment pattern allows rollback during production issues?

- A. Canary deployment
- B. Random deployment
- C. Manual deployment only
- D. FTP deployment

Correct Answer:

- A. Canary deployment
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Q43. Which security principle limits AI agent permissions to minimum required access?

- A. Least privilege
- B. Open access
- C. Shared admin access
- D. Public access

Correct Answer:

- A. Least privilege
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Q44. Which architecture pattern is BEST for enterprise AI scalability?

- A. Distributed microservices architecture
- B. Single desktop application
- C. Offline processing only
- D. USB-based architecture

Correct Answer:

- A. Distributed microservices architecture
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Q45. Which Azure capability helps securely store enterprise secrets and API keys?

- A. Azure Key Vault
- B. Notepad
- C. Device Manager
- D. Paint

Correct Answer:

- A. Azure Key Vault
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FINAL REVISION QUESTIONS

Q46. What is the PRIMARY purpose of AI orchestration?

- A. Delete databases
- B. Coordinate workflows between agents and systems
- C. Disable monitoring
- D. Increase hardware temperature

Correct Answer:

- B. Coordinate workflows between agents and systems
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Q47. Which Microsoft platform supports enterprise low-code AI application development?

- A. Power Platform
- B. BIOS
- C. DOS
- D. CMD

Correct Answer:

- A. Power Platform
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Q48. Which AI workflow step validates approval policies automatically?

- A. AI reasoning engine
- B. Disk defragmentation
- C. DHCP service
- D. DNS caching

Correct Answer:

- A. AI reasoning engine
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Q49. Which enterprise AI capability improves user personalization?

- A. Context retention
- B. Monitor brightness
- C. BIOS updates
- D. File compression

Correct Answer:

A. Context retention

Q50. Which Microsoft service supports enterprise semantic document retrieval?

- A. Azure AI Search
- B. MS Paint
- C. Remote Desktop
- D. DHCP Server

Correct Answer:

A. Azure AI Search

END OF PRACTICE TEST

Score Guide

Score	Readiness
90%+	Strongly Ready
80–89%	Ready
70–79%	Borderline
Below 70%	Needs More Practice